

Introduction to Deep Learning

Gunhee Kim



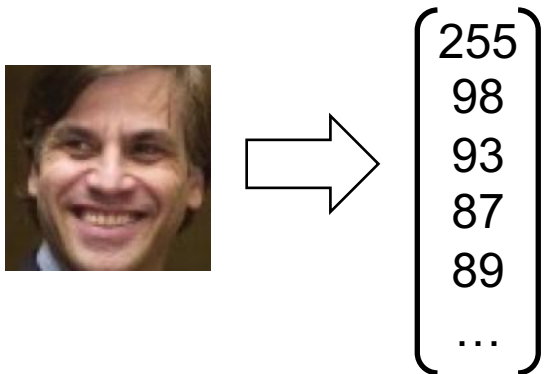
서울대학교
SEOUL NATIONAL UNIVERSITY

Deep Learning for Image Classification

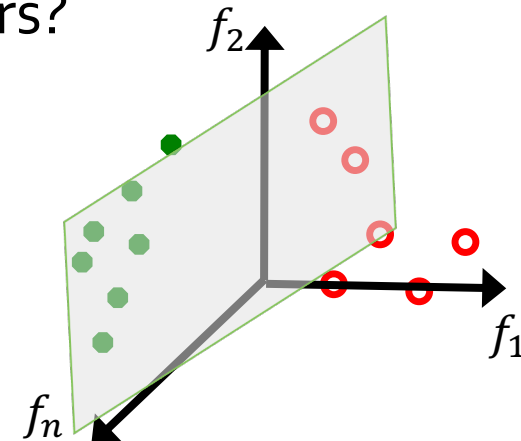
- Representation learning attempts to automatically learn good features or representations
- Feature learning problem
 - Suppose we want to classify images whether they are faces or not



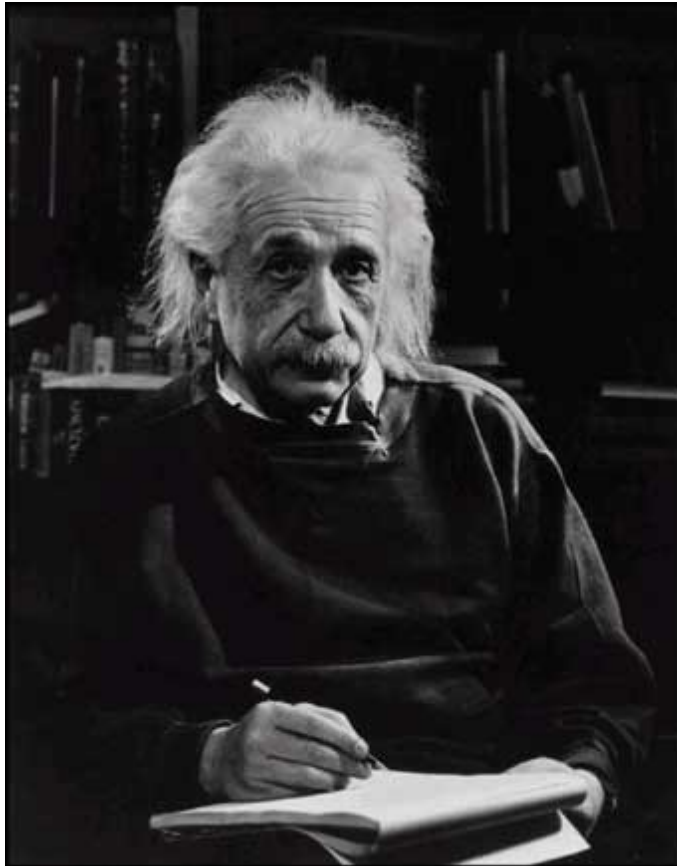
- Can we represent images using 100 real numbers?



Can we find
a good one?



Representation is Not Easy – An Image



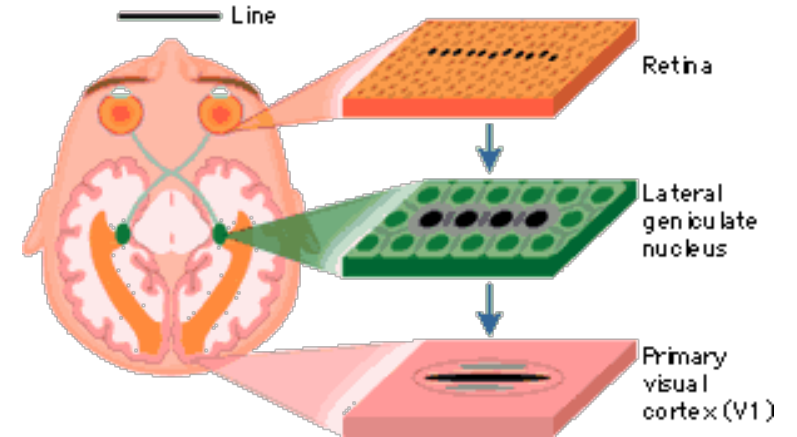
What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

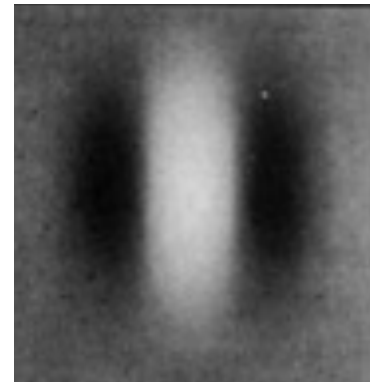
What a computer see

Edge Detection

- Our brain first detects edges
 - Cells in primary visual cortex (V1) are activated by lines of a given orientation
- First stage of visual processing: V1



Neuron #1 of visual cortex (model)



Neuron #2 of visual cortex (model)

Edge Detection

- Line segments where the image brightness changes sharply (or has discontinuities)

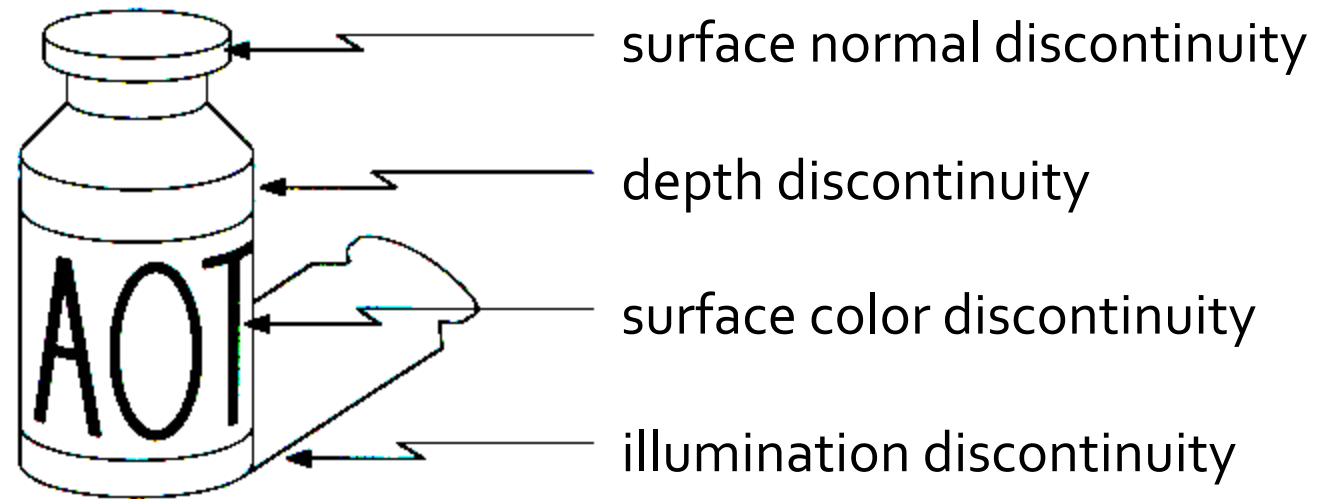


- Edge detection is NOT easy



Edge

- Edges are caused by a variety of factors



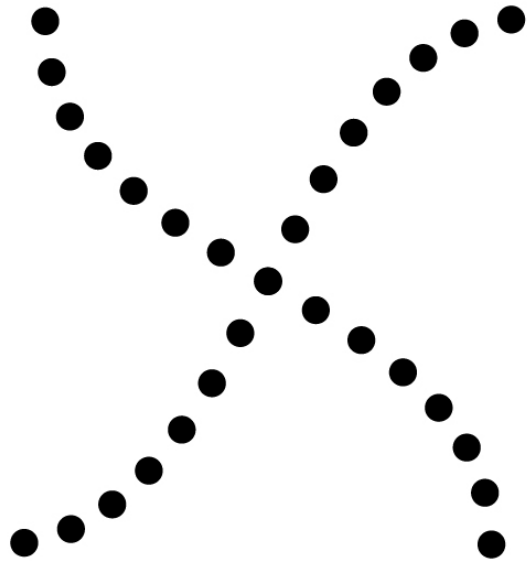
- OK, edge detection is not easy... then how can we group edges?

Grouping

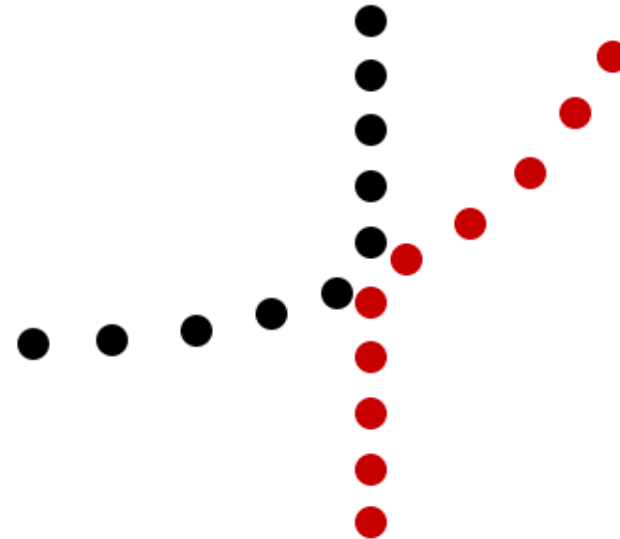


Grouping

- People adaptively use different rules for grouping



People tends to mentally
form a continuous line



All of sudden, people use
color information

Mid-Level Representation

- Mid-level cues



"Tokens" from Vision
by David Marr



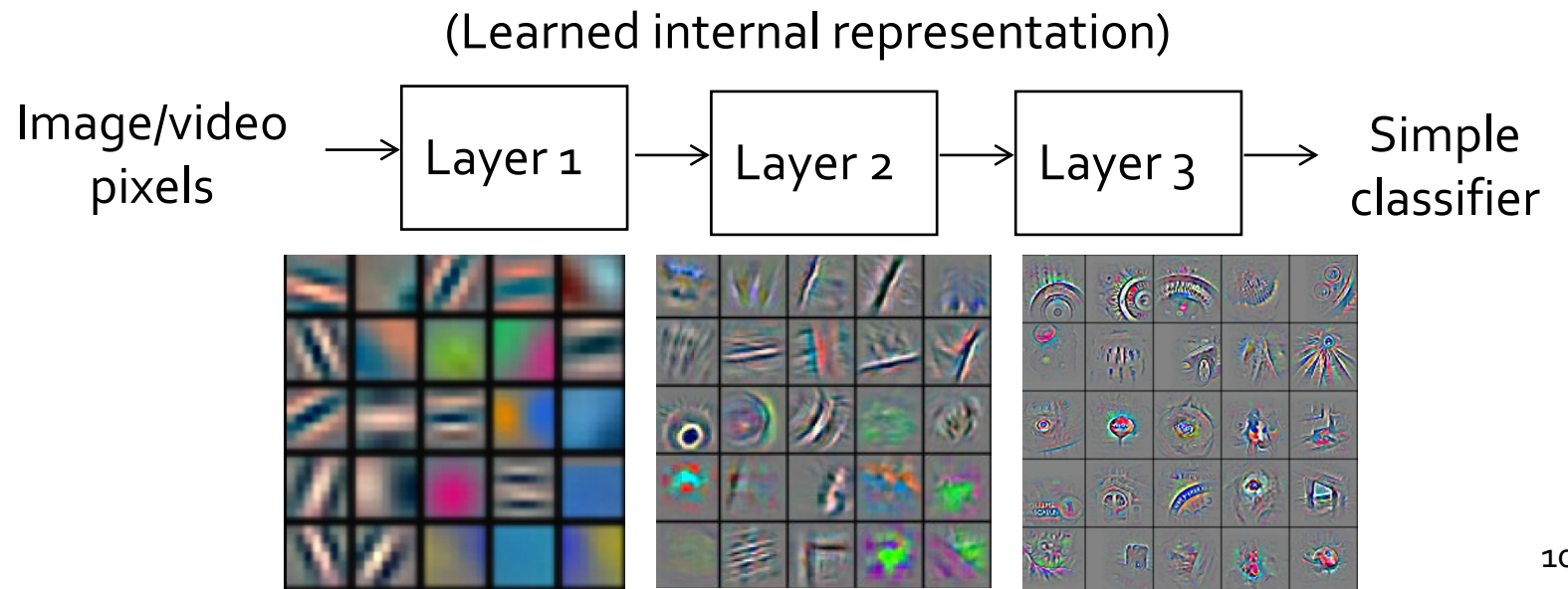
- Object parts



- Difficult to hand-engineer → What about learning them?

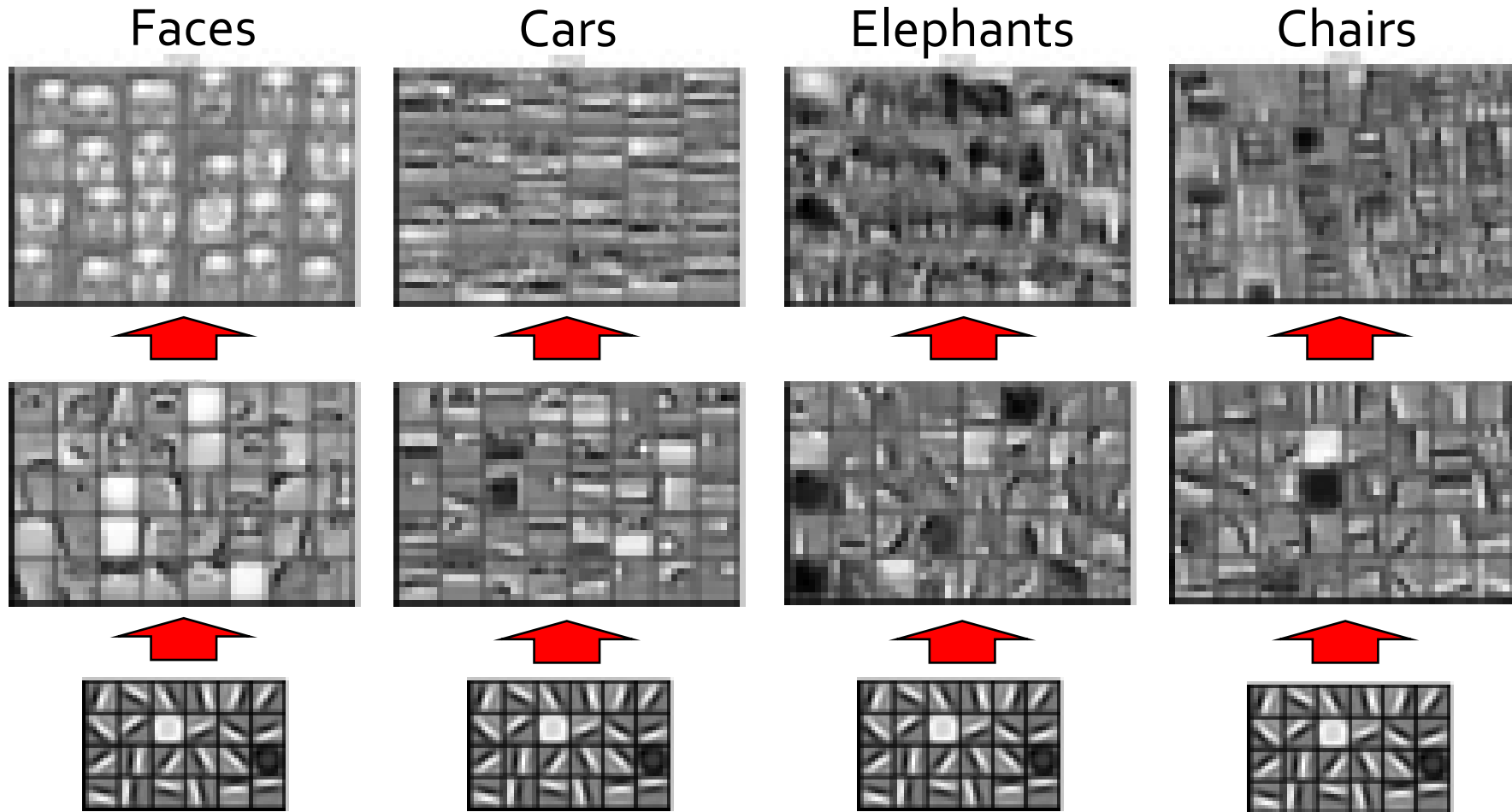
Deep Learning

- Deep learning algorithms attempt to learn multiple levels of representation of increasing complexity abstraction
 - A cascade of many layers of nonlinear processing units for feature extraction and transformation
 - Each hidden layer learns different level of abstraction; the levels form a hierarchy of concepts
 - End-to-end: All the way from pixels → Classifier



Learning of Object Parts

- Examples of learned object parts from object categories



Traditional Pattern Recognition

Vision



Features
(fixed)

SIFT/HOG

Representation
(unsupervised)

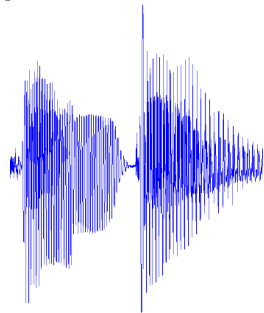
K-means /
pooling

(supervised)

Classifier

Car

Speech



MFCC

Mixture of
Gaussians

Classifier

word

NLP

This burrito place is
yummy and fun!

Parse tree

N-grams

Classifier

POS

(Deep) Hierarchical Compositionality

Vision

Pixels → Edges → Texton → Motif → Part → Object

Speech

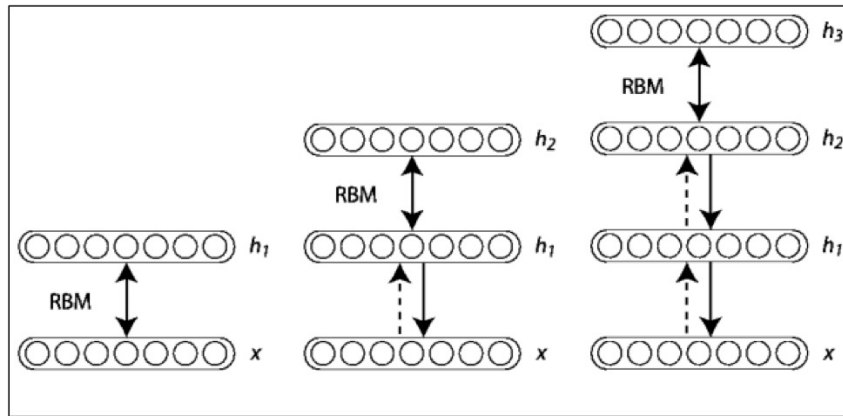
Sample → Spectral → Formant. → Motif → Phone. → Word
band

Natural language

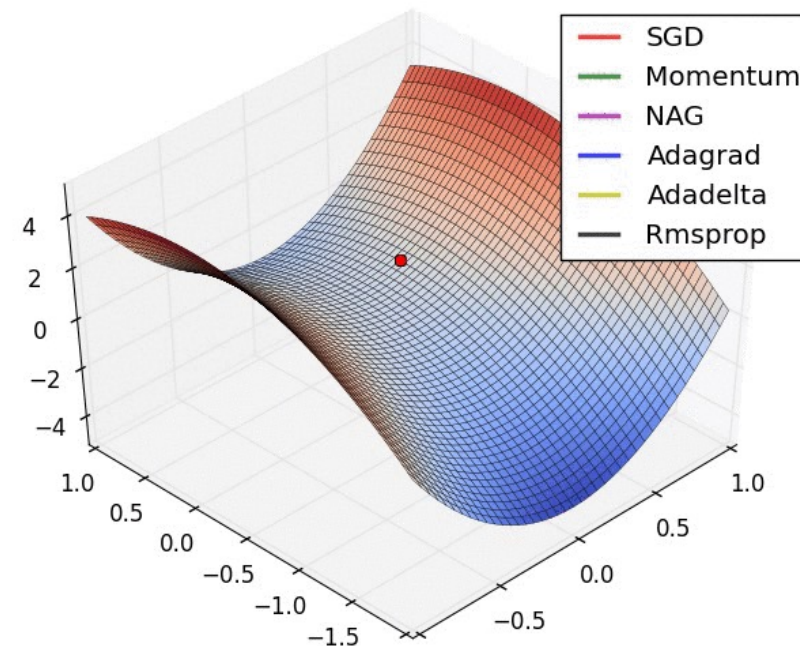
Character → Word → NP/VP/... → Clause → Sentence → Story

How Can DL Revive?

- Progress in machine learning research
 - Before 2006 training deep architectures was unsuccessful
 - New methods for unsupervised pre-training have been developed (RBMs, autoencoders, contrastive estimation, etc)
 - More efficient parameter estimation methods
 - Better understanding of model regulariz

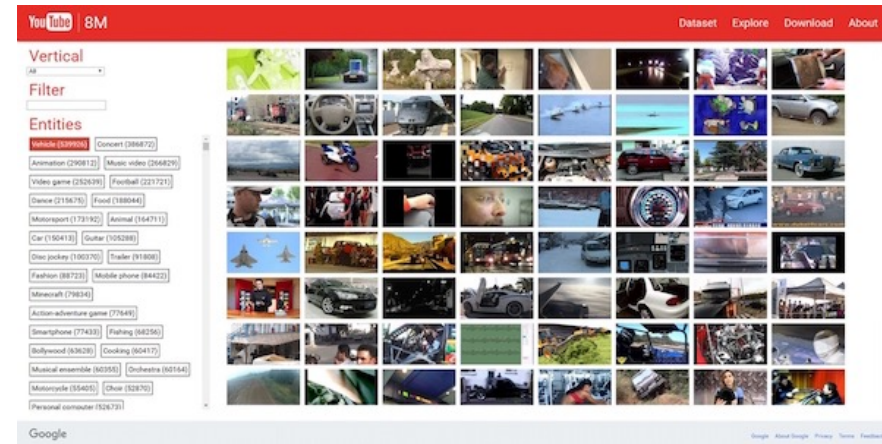


[Hinton et al. 2006]



How Can DL Revive?

- Many training data available

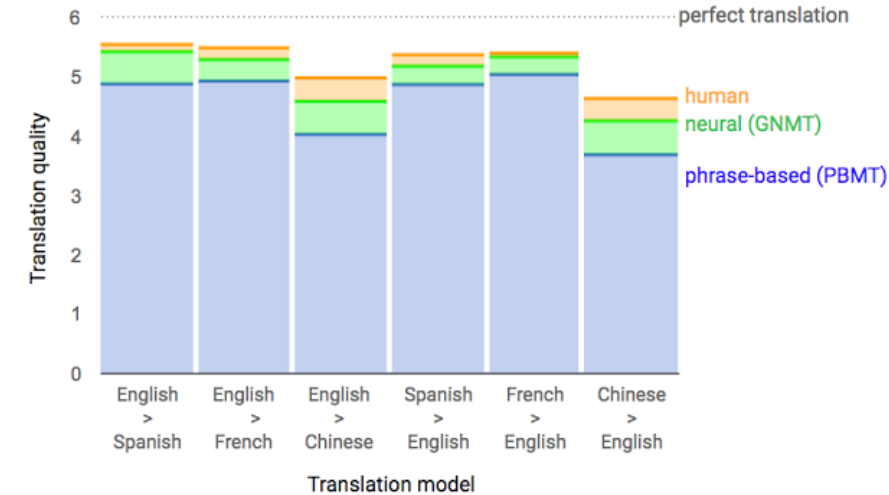


- Changes in computing technology favor deep learning
 - Multi-core CPUs and GPUs
 - Uniform parallel operations on dense vectors are faster



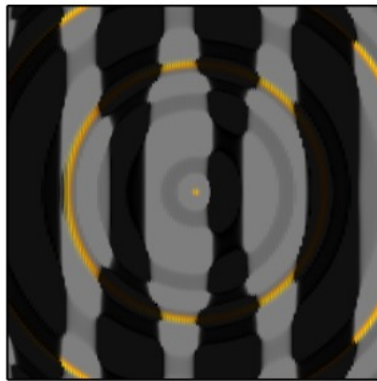
Limitation – 1. Need Many Clean Training Data

- Machine translation is so successful.
Then how are about the other NLP tasks?
 - Google Neural Machine Translation system (GNMT) in 2016/09
- Spell checking
 - In Google News, 곱배기 (254 results) vs 곱빼기 (683)
 - 외래어 표기법: 루이비통 vs 루이뷔통, 마를린 먼로 vs 메릴린 먼로
- Sentiment analysis
 - Dorothy Parker on Katherine Hepburn:
"She runs the gamut of emotions from A to B"



Limitation – 2. Easily Break Down

- Deep neural networks are easily fooled
 - High confidence predictions for unrecognizable images
 - State-of-the-art DNNs trained on ImageNet believe with $\geq 99.6\%$ certainty to be a familiar object



king penguin



starfish



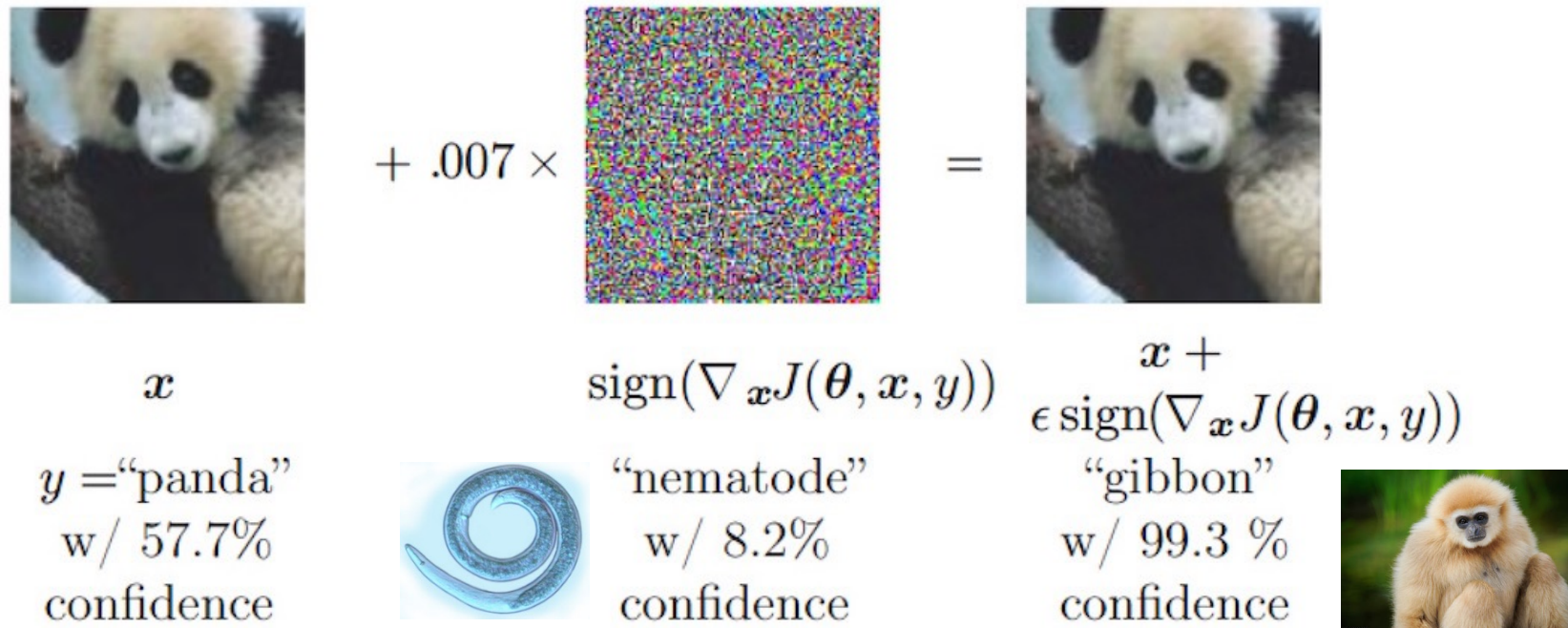
armadillo



School bus!

Limitation – 2. Easily Break Down

- Adversarial examples
 - Human cannot tell the difference with the original example
 - However, the network can make highly different predictions
 - e.g., Fast Gradient Sign Method (FGSM)



Limitation – 3. Not Energy Efficient

- Not sustainable energy consumption in Nature
 - Lee Sedol used about **20 Watts** of power to operate
 - AlphaGo used approximately **1 MW** (200 W per CPU and 200 W per GPU)

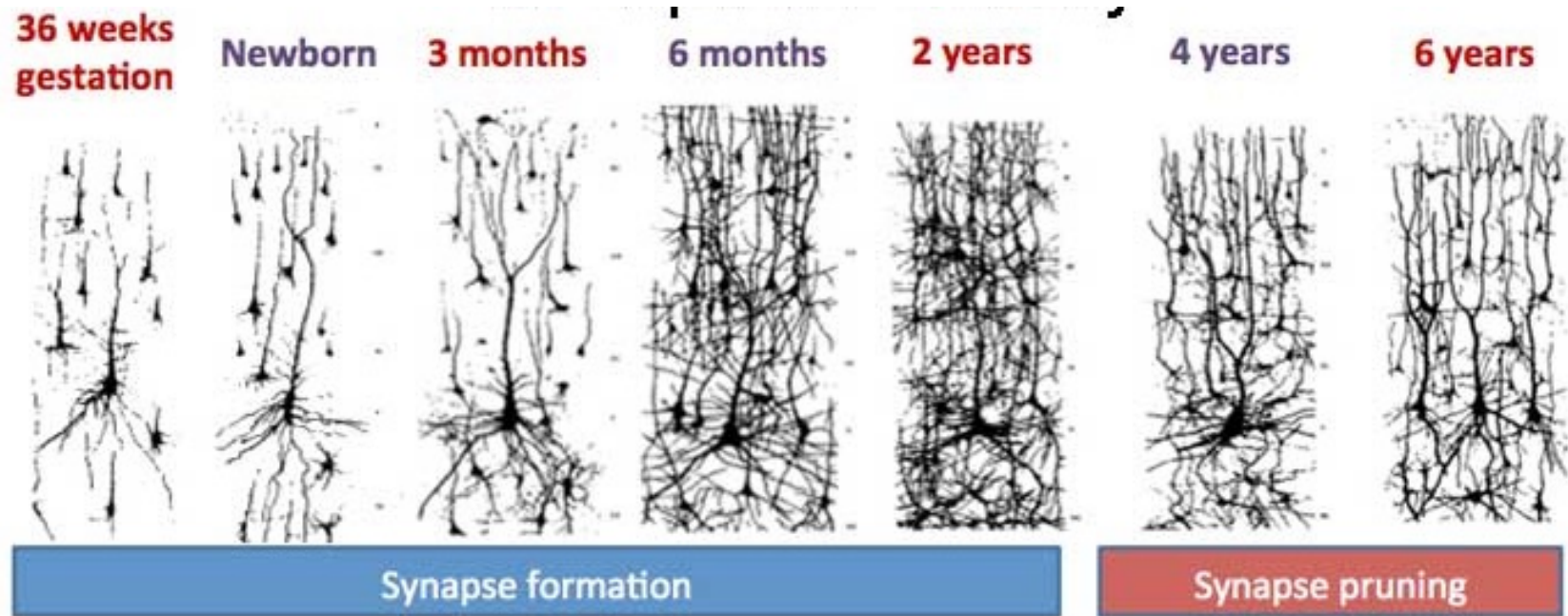
50,000
times
more!



AlphaGO	Lee Se-dol
1202 CPUs, 176 GPUs, 100+ Scientists.	1 Human Brain, 1 Coffee.

Limitation – 3. Not Energy Efficient

- Doing nothing is often the best action!
- Developmental plasticity
 - Each neuron in the cerebral cortex has approximately...
 - 2,500 synapses at birth → 15,000 synapses → keep decreasing in our entire world



Limitation – 4. Lack of Semantic Information

- Human can learn a new class even with a single image
 - Suppose my kid knows jaguar, and leopard, and see a picture of cheetah for the first time



Does it have a **tail**?

Does it lay the **egg**?

How does its **foot** look like?

- Generalization / Specialization
 - First do categorization by finding commonality (it's a big cat)
 - Then focus on its differences in the group (e.g. tear marks, patterns, ears, ...)

Limitation – 4. Lack of Semantic Information

- DL models require a large amount of training data
 - Knowledge transfer is difficult
 - Collect training data of a new class again...



1,000 images
of Jaguar



1,000 images
of Leopard



1,000 images
of Cheetah

- Promising research directions
 - Zero-shot/one-shot learning, transfer learning, multi-task learning, semi-/unsupervised learning... g. tear marks, patterns, ears, ...)

Limitation – 5. Interpretability/Explainability

- Deep networks are widely regarded as black boxes but are often more accurate
 - State-of-the-art CNNs often include 10~100 millions of parameters to learn
 - It is hard to know what happens inside the model

